Timeline and deliverables for Team Project Part 2

You are expected to use the engine you built over the first 7 weeks of this module to create one of the following:

- **Game:** It should have some engaging gameplay, designed aligned to a real-world expectations or requirements (See samples below)
- **Simulation:** It should simulate something real world (see samples below)

The problem must be solved well, so the client (based on the selected problem and lab in charge approval) is happy and it would address the target audience's needs.

Remember the focus for this project is on the **quality of the code** your product and your ability to take **reusable components** (your engine) to solve a separate problem (your game or simulation), so the complexity of the game or simulation is not the focus of your grade, find a way to keep these things interesting but simple while still demonstrating the capabilities of your engine and showcasing the quality of your code.

While developing your game or simulation, continue to:

- Adhere to software design principles such as **SOLID**
- Implement at least **three design patterns**
- Ensure a clear separation between engine code (more abstract side of things / managers) and game/simulation code (more concrete side of things specific to your game/simulation), to demonstrate appropriate architecture.

Some suggestions for a game that would algin to real-world expectation or requirement might include:

- Getting kids interested in **Language Learning**
- Educating families and parents about Healthy Eating
- Getting families aware of and supporting **Sustainable Living**
- **Your own idea to be discussed with your lab-in-charge**

Some suggestions for a simulation that may simulate something in the real world:

- A rocket simulation
- A crowd simulator
- A traffic simulator
- A physics sandbox (example: simple suspension simulation or solar system gravity simulation)
- **Your own idea to be discussed with your lab-in-charge**

Continue to appropriately use classes, objects, inheritance, polymorphism and error handling

Ensure your code is robust, reusable and most importantly scalable, for example, is it easy to add more players, more enemies, more levels, more menus (for a game), or is it easy to add more planets, more rockets, more cars (for the simulation examples above). Does your code have low coupling? Would it be easy to use classes and components you've created in other applications without needing to change much or import irrelevant classes?

The application of good object-oriented programming practices to solve a problem is the main focus here, the game design/simulation complexity won't be graded apart from 1. You've come up with an interesting problem to solve and solved it and 2. how good it is at showcasing your engine and your

mastery of using reusable code, writing good code, object-oriented programming, design patterns and good architecture design.

Submission Details

Final Submission: **Friday 27th March 2026, 23:59**

You are expected to submit a report, presentation and video, peer evaluation and self-reflection as well as a working project code (Engine and Game/Simulation), 10 minute video for the presentation, details below.

Each Group is to submit a zip file to their groups xSITE Dropbox ('Team Project Part 2').

Please follow the naming convention carefully when submitting your project and structuring your folders.

Zip File Name:

- [TeamNumber_ProjectPart2]

Zip File Folder Structure and Content:

- Report (your report and any supporting documents)
- Presentation (slides for your 10 minute presentation)
    - Not required to present physically, video presentation is expected, details below
- Project (all project and code files for your final project deliverable)
- Reflection (link will be sent on week 12)
- Txt file with presentation video link, any other links which may be useful to assessment of your work

One Submission Only per team. If you wish to resubmit before the deadline, only the latest submission will be graded and all other submissions will be ignored.

Final Report

Final report should not be more than 15 pages and contain the following:

- Overall system design for the game engine and the game
    - Overview/Purpose of Components and Layers used
    - Discussion of Design patterns implemented
    - Improvements to the engine since last submission
- Complete UML diagram for the Engine and the Game or Simulation
- Object oriented principles adopted in the system and justification
- Reflection on limitations with the Engine, Game or Simulation and Software Design
- Clear and concise description of each team member's contribution.
    - GAI-Teammate is a team member, you must also reflect on their contribution
- NOTE: EVERY team member must contribute to the implementation
- Things that you want to highlight, for example unique and innovative features in your game engine and game (game mechanics)

Presentation and Video

You will not be required to present your project physically, but are expected to create a ***concise 10 minute video*** which meets the below expectations:

Deck of slides: 10 – 12 slides

The deck of slides should:

- Highlight the key aspects of the game engine (consider in terms of scalability, reusability, and design)
- Highlight key improvements to the engine based on your previous submission
- Exhibit abstract engine implementation, showcasing the implementation of the must haves using your game or simulation to do so.
- Discuss your game and how it addresses the proposed problem that either you have derived and cleared with your lab in charge or related to one of the topics proposed in listed in this document.
- Demo the game while discussing the above
- Be comprehensive and self-explanatory
- Summarize your report

Peer Evaluation and Self-Reflection

You will be asked to complete a peer evaluation as well as a self-reflection. These will be sent to you closer to the deadline of your final project deliverables.